THE COMPLETE DATA ANALYST BOOTCAMP

Miramore Institute | Learn. Create. Lead.

COURSE 1 — INTRODUCTION TO DATA ANALYSIS

6 Lessons

- 1.1 What is data analysis and why it matters
- 1.2 The data analyst role — responsibilities and career path
- 1.3 The data analysis workflow — from raw data to decision
- 1.4 Types of data — structured, unstructured, quantitative, qualitative
- 1.5 Data ethics, privacy and bias — what every analyst must know
- 1.6 Hands-on — download and explore your first real dataset

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COURSE 2 — MICROSOFT EXCEL FOR DATA ANALYSIS

28 Lessons

Section 1 — Excel Interface and Essentials

- 2.1 Navigating the Excel interface — ribbon, sheets and shortcuts
- 2.2 Formatting cells, numbers and tables for clean reporting
- 2.3 Freeze panes, filters and sorting — working with large datasets

Section 2 — Formulas and Functions

- 2.4 Essential math functions — SUM, COUNT, AVERAGE, SUBTOTAL
- 2.5 Logical functions — IF, AND, OR and nested IF statements
- 2.6 Conditional aggregation — SUMIF, COUNTIF, SUMIFS, COUNTIFS
- 2.7 Text functions — TRIM, LEFT, RIGHT, CONCAT, TEXT, LEN
- 2.8 Lookup functions — VLOOKUP, HLOOKUP, INDEX MATCH
- 2.9 Modern lookups — XLOOKUP and dynamic array functions
- 2.10 Error handling — IFERROR, ISNUMBER and data validation

Section 3 — Data Cleaning and Transformation

- 2.11 Identifying and removing duplicates from a dataset
- 2.12 Handling missing data — techniques and best practices
- 2.13 Standardizing inconsistent data — dates, names and categories

Section 4 — Statistical Analysis

- 2.14 Descriptive statistics — MAX, MIN, MEDIAN, MODE, STDEV, VARIANCE
- 2.15 AVERAGEIF and AVERAGEIFS for conditional averages
- 2.16 Percentile, rank and frequency distribution analysis
- 2.17 Trend analysis and basic regression in Excel

Section 5 — Charts and Visualization

- 2.18 Choosing the right chart — bar, line, pie, scatter, combo
- 2.19 Customizing charts for professional reports
- 2.20 Conditional formatting as a visual analysis tool

Section 6 — Pivot Tables and Dashboards

- 2.21 Building pivot tables from scratch — grouping, filtering, sorting
- 2.22 Calculated fields and measures in pivot tables
- 2.23 Pivot charts — linking visuals to pivot data
- 2.24 Building an interactive dashboard with slicers and timelines
- 2.25 Dashboard design principles — layout, color and storytelling

Section 7 — Capstone Projects

- 2.26 Project — Build a complete sales performance dashboard from raw data
- 2.27 Project — Customer analysis using pivot tables and charts
- 2.28 Project walkthrough and assignment brief

COURSE 3 — SQL FOR DATA ANALYSIS

30 Lessons

Section 1 — Setup and Databases

- 3.1 Installing MySQL and setting up MySQL Workbench
- 3.2 What is a database — tables, rows, columns and relationships
- 3.3 RDBMS concepts — primary keys, foreign keys and normalization
- 3.4 Loading a sample database and exploring it in Workbench

Section 2 — Core SQL Queries

- 3.5 SELECT, FROM, WHERE — writing your first queries
- 3.6 Filtering data — AND, OR, IN, NOT IN, BETWEEN, LIKE
- 3.7 Sorting and limiting results — ORDER BY and LIMIT
- 3.8 Aggregating data — SUM, AVG, COUNT, MIN, MAX
- 3.9 Grouping and filtering groups — GROUP BY and HAVING

Section 3 — Working with Multiple Tables

- 3.10 Understanding joins — INNER JOIN with real examples
- 3.11 LEFT JOIN and RIGHT JOIN — including unmatched records
- 3.12 Aliases — making complex queries readable
- 3.13 Subqueries — writing a query inside a query
- 3.14 EXISTS vs IN — when to use each
- 3.15 CTEs — WITH clause for clean readable analysis

Section 4 — SQL Functions

- 3.16 String functions — CONCAT, UPPER, LOWER, SUBSTRING, TRIM
- 3.17 Numeric functions — ROUND, FLOOR, CEIL, MOD, ABS
- 3.18 Date and time functions — DATEDIFF, DATE FORMAT, NOW, YEAR, MONTH
- 3.19 CASE statements — adding conditional logic to queries
- 3.20 Window functions — ROW NUMBER, RANK, DENSE RANK, LAG, LEAD

Section 5 — Data Management

- 3.21 Creating tables — CREATE TABLE, data types and constraints
- 3.22 Inserting, updating and deleting data safely
- 3.23 Handling duplicate records — identifying and removing duplicates
- 3.24 Creating and using views for reusable analysis
- 3.25 Indexing for query performance optimization

Section 6 — Automation and Capstone

- 3.26 Stored procedures — writing reusable SQL logic
- 3.27 SQL triggers — automating actions on data changes
- 3.28 Project — Employee performance analysis using SQL
- 3.29 Project — E-commerce sales analysis using joins, aggregations and ranking
- 3.30 Project walkthrough and assignment brief

COURSE 4 — POWER BI FOR DATA ANALYSIS

32 Lessons

Section 1 — Connecting and Transforming Data

- 4.1 Power BI interface tour — report, data and model views
- 4.2 Connecting to data sources — Excel, CSV, SQL Server and web
- 4.3 Power Query editor — transforming columns, renaming and changing types
- 4.4 Filtering rows and removing unnecessary columns
- 4.5 Splitting columns, replacing values and handling nulls
- 4.6 Merging and appending queries — combining multiple tables
- 4.7 Pivoting and unpivoting data for the right analysis shape
- 4.8 Grouping and aggregating in Power Query

Section 2 — Building the Data Model

- 4.9 Star schema vs snowflake schema — which to use and why
- 4.10 Fact tables vs dimension tables — building a clean model
- 4.11 Creating and managing relationships between tables
- 4.12 Cardinality, cross filter direction and active vs inactive relationships

Section 3 — DAX for Data Analysis

- 4.13 What is DAX — calculated columns vs measures explained
- 4.14 Row context vs filter context — the most important DAX concept
- 4.15 Essential aggregation measures — SUM, AVERAGE, COUNT, DISTINCTCOUNT
- 4.16 Logical and conditional DAX — IF, SWITCH, AND, OR
- 4.17 CALCULATE — the most powerful DAX function
- 4.18 FILTER and ALL — controlling and removing filters
- 4.19 Time intelligence — MTD, QTD, YTD and year over year comparisons
- 4.20 Iterator functions — SUMX, AVERAGEX, RANKX
- 4.21 RELATED and LOOKUPVALUE — joining data across tables with DAX

Section 4 — Reports and Visualizations

- 4.22 Adding and formatting visuals — bar, line, pie, card and KPI
- 4.23 Matrix visual — the pivot table of Power BI
- 4.24 Maps — basic map, filled map and bubble map
- 4.25 Slicers, timelines and report level filters for interactivity
- 4.26 Editing visual interactions — controlling how visuals affect each other
- 4.27 Drillthrough and drill down for deeper data exploration
- 4.28 Bookmarks and what if parameters for dynamic reports
- 4.29 Row level security — controlling who sees what data

Section 5 — Capstone Projects

- 4.30 Project — End to end sales performance dashboard from model to report
- 4.31 Project — HR analytics dashboard — headcount, attrition and salary bands
- 4.32 Project walkthrough and assignment brief

COURSE 5 — TABLEAU FOR DATA ANALYSIS

28 Lessons

Section 1 — Getting Started with Tableau

- 5.1 Tableau interface tour — sheets, dashboards and stories
- 5.2 Connecting to data — Excel, CSV and SQL Server
- 5.3 Dimensions vs measures and discrete vs continuous explained
- 5.4 Joining and blending data from multiple sources

Section 2 — Building Charts and Reports

- 5.5 Your first Tableau chart — drag, drop and Show Me
- 5.6 Bar charts — grouped, stacked and side by side
- 5.7 Line charts and dual axis charts for trends and comparisons
- 5.8 Scatter plots and bubble charts for correlation analysis
- 5.9 Heatmaps and highlight tables for pattern detection
- 5.10 Maps — basic geographic map and filled map
- 5.11 Tree map, pie chart and box plot — when to use each
- 5.12 Waterfall chart and Gantt chart for business reporting
- 5.13 Adding labels, tooltips and reference lines to charts

Section 3 — Organizing and Segmenting Data

- 5.14 Grouping data and creating bins for segmentation
- 5.15 Sets and combined sets for dynamic comparisons
- 5.16 Hierarchies — drilling from year down to day

- 5.17 Sorting, totals, subtotals and grand totals in crosstabs

Section 4 — Calculations and Filters

- 5.18 Creating calculated fields — basic formulas and IF logic
- 5.19 Table calculations — running totals, percent of total and ranking
- 5.20 Parameters — letting users control the analysis
- 5.21 Filters — dimension, measure, context and top bottom filters

Section 5 — Dashboards and Stories

- 5.22 Building a dashboard — layout, containers and objects
- 5.23 Dashboard actions — filter, highlight and URL actions
- 5.24 Device specific layouts — designing for desktop and mobile
- 5.25 Storytelling with data — building a Tableau story

Section 6 — Capstone Projects

- 5.26 Project — Retail sales dashboard — regional trends and KPIs
- 5.27 Project — Public health data visualization — disease and population trends
- 5.28 Project walkthrough and assignment brief

COURSE 6 — PYTHON FOR DATA ANALYSIS

38 Lessons

Section 1 — Python Fundamentals

- 6.1 Installing Anaconda, Jupyter Notebook and VS Code
- 6.2 Variables, data types and operators in Python
- 6.3 Control flow — if else conditions and for while loops
- 6.4 Functions — writing reusable code blocks
- 6.5 Lambda functions and list comprehensions for concise code
- 6.6 Error handling — try except for robust scripts
- 6.7 Python data structures — lists, tuples, sets and dictionaries

Section 2 — NumPy for Numerical Computing

- 6.8 Creating and working with NumPy arrays
- 6.9 Array indexing, slicing and reshaping
- 6.10 Element wise operations and broadcasting
- 6.11 Statistical functions with NumPy — mean, std, sum, percentile

Section 3 — Pandas for Data Manipulation

- 6.12 Pandas Series and DataFrames — the core data structures
- 6.13 Reading and writing CSV, Excel and JSON files
- 6.14 Exploring a dataset — head, info, describe, shape, dtypes
- 6.15 Filtering and selecting data — loc, iloc and conditions
- 6.16 Handling missing data — detecting, filling and dropping nulls
- 6.17 Removing duplicates and standardizing data types
- 6.18 Transforming columns with apply and map
- 6.19 Sorting, grouping and aggregating data
- 6.20 Merging and joining DataFrames — merge and concat
- 6.21 Pivot tables in Pandas — summarizing data quickly

Section 4 — Data Visualization

- 6.22 Matplotlib basics — line, bar and scatter plots
- 6.23 Customizing Matplotlib — titles, labels, colors and subplots
- 6.24 Seaborn for statistical visualization — heatmaps, boxplots, pairplots
- 6.25 Choosing the right chart and storytelling with data

Section 5 — Statistical Analysis

- 6.26 Descriptive statistics — mean, median, mode, variance, standard deviation
- 6.27 Probability distributions — normal, binomial, Poisson
- 6.28 Hypothesis testing — t-test and chi-square test with SciPy
- 6.29 Correlation and regression analysis

- 6.30 Confidence intervals and statistical inference

Section 6 — Object Oriented Programming for Analysts

- 6.31 Classes and objects — why OOP matters for data work
- 6.32 Inheritance and encapsulation — building reusable data tools
- 6.33 Project — Build a data cleaning class for any dataset

Section 7 — Capstone Projects

- 6.34 Project — Full data analysis pipeline from raw CSV to insights
- 6.35 Project — Customer segmentation analysis with Pandas and Seaborn
- 6.36 Project — Financial fraud detection — exploratory data analysis
- 6.37 Final project — End to end multi source analytics dashboard in Python
- 6.38 Project walkthrough and assignment brief

COURSE SUMMARY

Course	**Tool**	**Lessons**
Course 1	Introduction to Data Analysis	6 lessons
Course 2	Microsoft Excel	28 lessons
Course 3	SQL	30 lessons
Course 4	Power BI	32 lessons
Course 5	Tableau	28 lessons
Course 6	Python	38 lessons
Total	**6 Tools**	**162 lessons**

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